

Eric A. Chadwick

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Education

- May., 2020 – Jun., 2025 **PhD, University of Toronto** Mechanical & Industrial Engineering
Thesis title: *Microscale Transport in Polymer Electrolyte Membrane Fuel Cells: Impact of Gravity and Flow Field Design.*
Supervisors: Prof. Aimy Bazylak & Prof. Dr. Volker P. Schulz
- May, 2017 – Sep., 2019 **MASc, University of Toronto** Mechanical & Industrial Engineering
Thesis title: *Vessel Network Extraction via micro-CT Imaging: A Foundation for Modelling Lung De- and Recellularization*
Supervisor: Prof. Aimy Bazylak
- Sep., 2012 – Apr., 2017 **BEng, University of Guelph** Mechanical Engineering
Thesis title: *Smart Thermal Battery System*
Supervisor: Prof. Shohel Mahmud

Employment History

- Nov., 2019 – Present **Academic Staff & Researcher**, Baden-Württemberg Cooperative State University
Courses Taught: *Numerical Methods for Engineers, Introduction to Python, Transport in Porous Media*
- May, 2020 – Jan., 2025 **Research Assistant (PhD)**, University of Toronto
- Sept. – Oct., 2020 **Lab Demonstrator Teaching Assistant**, University of Toronto
Course: *Mechanical & Thermal Energy Conversion*
- Jan. – Apr., 2018 **Head Tutorial Teaching Assistant**, University of Toronto
Course: *Heat & Mass Transport*
- May, 2017 – Sep., 2019 **Research Assistant (MASc)**, University of Toronto
- Jan. – Aug., 2016 **Undergraduate Research Assistant**, University of Guelph
Research Topic: *Multiphase flow and efficiency analysis of an air lift pump*
- May, 2015 – Aug., 2015 **Quality Engineering Intern**, Ontario Power Generation
- May, 2014 – Apr., 2015 **Undergraduate Research Assistant**, University of Guelph
Research Topic: *Minimum quantity lubrication strategies for the machining of hard-to-cut materials*

Research Publications

Journal Articles

- 1 Y. Dumay, **E.A. Chadwick**, M. Napoléon, C. Seide, B. Ortlepp, N. Blessing, and V. P. Schulz, "Development of an additively manufactured vapor chamber using hybrid wick structures", *Applied Thermal Engineering*, p. 126 442, 2025. [DOI: 10.1016/j.applthermaleng.2025.126442](https://doi.org/10.1016/j.applthermaleng.2025.126442)
- 2 **E.A. Chadwick**, B. Derebaşı, V. P. Schulz, and A. Bazylak, "Influence of gravity on water management in polymer electrolyte membrane fuel cells", *Scientific Reports, Special Issue: Transport Phenomena in Electrochemical Devices*, vol. 15, p. 39 380, 2025. [DOI: 10.1038/s41598-025-09067-y](https://doi.org/10.1038/s41598-025-09067-y)

- 3 E.A. Chadwick, P. Shrestha, H. B. Parmar, A. Bazylak, and V. P. Schulz, "Biomimetic auxiliary channels enhance oxygen delivery and water removal in polymer electrolyte membrane fuel cells", *Applied Energy*, vol. 389, p. 125760, 2025.  DOI: 10.1016/j.apenergy.2025.125760
- 4 D. Parekh, S. Ranieri, T. Seip, E.A. Chadwick, B. Derebaşı, N. Ge, R. Wang, and A. Bazylak, "Dominating impact of microporous layer thickness on gas diffusion layer oxygen transport resistance", *Journal of Power Sources*, vol. 656, p. 238031, 2025.  DOI: 10.1016/j.jpowsour.2025.238031
- 5 J. K. Chan, E.A. Chadwick, T. Taniguchi, A. Ahmadipour, T. Suzuki, D. Romero, C. Amon, T. Waddell, G. Karoubi, and A. Bazylak, "Predicting cell distributions in lung vasculature to optimize re-endothelialization", *Frontiers in Bioengineering and Biotechnology*, vol. 10, pp. 2296–418, 2022.  DOI: 10.3389/fbioe.2022.891407
- 6 E.A. Chadwick, L. Hammen, V. P. Schulz, A. Bazylak, M. Ioannidis, and J. Gostick, "Incorporating the effect of gravity into image-based drainage simulations on volumetric images of porous media", *Water Resources Research*, vol. 58, e2021WR031509, 2022.  DOI: 10.1029/2021WR031509
- 7 E.A. Chadwick, T. Suzuki, M. George, D. Romero, C. Amon, T. Waddell, G. Karoubi, and A. Bazylak, "Vessel network extraction and analysis of mouse pulmonary vasculature via x-ray micro-computed tomographic imaging", *PLOS One Computational Biology*, vol. 17, e1008930, 2021.  DOI: 10.1371/journal.pcbi.1008930

Conference Proceedings

- 1 R. Abed, E.A. Chadwick, and W. Ahmed, "Two-phase flow behaviour in airlift pumps", in *Proceedings of the 5th International Conference of Fluid Flow, Heat and Mass Transfer*, Niagara Falls, Canada, 2018, pp. 7–9.  URL: https://avestia.com/FFHMT2018_Proceedings/files/paper/FFHMT_168.pdf

Conference Presentations

- 1 E.A. Chadwick, T. Dörenkamp, J. Eller, V. Schulz, and A. Bazylak, *Operando μ ct analysis of porous flow fields for improved liquid water management in polymer electrolyte membrane fuel cells (best poster award winner)*, Presented at 76th Annual Meeting of the International Society of Electrochemistry, Mainz, Germany, 2025.
- 2 E.A. Chadwick, A. Briand, T. Seip, B. Auvity, A. Bazylak, and V. Schulz, *Balancing performance and pressure drop in ni foam flow fields via embedded channel design*, Presented at PRiME 2024, Honolulu, USA, 2024.  URL: <https://iopscience.iop.org/article/10.1149/MA2024-02443086mtgabs/meta>
- 3 E.A. Chadwick, B. Derebaşı, V. P. Schulz, and A. Bazylak, *Gravity's effect on two-phase transport in pem fuel cells: An operando synchrotron radiography investigation (2nd place poster prize)*, Presented at the Canadian Light Source Annual User Meeting, Waterloo, Canada, 2024.
- 4 E.A. Chadwick, V. Schulz, and A. Bazylak, *Effects of gravity on water transport in pem fuel cells via synchrotron x-ray radiography*, Presented at PRiME 2024, Honolulu, USA, 2024.  URL: <https://iopscience.iop.org/article/10.1149/MA2024-02443076mtgabs/meta>
- 5 E.A. Chadwick, P. Shrestha, H. Parmar, V. Schulz, and A. Bazylak, *Biomimetic auxiliary channels for enhanced pem fuel cell performance (general student poster award winner, 3rd place)*, Presented at PRiME 2024, Honolulu, USA, 2024.  URL: <https://iopscience.iop.org/article/10.1149/MA2024-02674661mtgabs/meta>
- 6 T. Seip, J. Lee, L. Zhu, S. Ranieri, E.A. Chadwick, K. Krause, L. Helfen, B. Auvity, S. Chevalier, and A. Bazylak, *Elucidating 3d water distributions in pem water electrolyzers via neutron imaging*, Presented at PRiME 2024, Honolulu, USA, 2024.  URL: <https://iopscience.iop.org/article/10.1149/MA2024-02453103mtgabs/meta>

- 7 L. Zhu, J. Lee, C. Ham, **E.A. Chadwick**, H. Parmar, P. Tritk, and A. Bazylak, *Compression of porous transport layers enhanced bubble transport in polymer electrolyte membrane water electrolyzers*, Presented at PRiME 2024, Honolulu, USA, 2024. [URL:](https://iopscience.iop.org/article/10.1149/MA2024-02453141mtgabs/meta)
<https://iopscience.iop.org/article/10.1149/MA2024-02453141mtgabs/meta>
- 8 **E.A. Chadwick**, H. Parmar, P. Shrestha, A. Bazylak, and V. Schulz, *Improving water management and reactant distribution in pem fuel cells via flow fields with biomimetic auxiliary channels*, Presented at ECS 244, Gothenburg, Sweden, 2023. [URL:](https://iopscience.iop.org/article/10.1149/MA2023-02381859mtgabs/meta)
<https://iopscience.iop.org/article/10.1149/MA2023-02381859mtgabs/meta>
- 9 H. Parmar, **E.A. Chadwick**, P. Shrestha, and A. Bazylak, *Tailoring parallel channel flow fields for efficient mass transport and compression in proton exchange membrane fuel cells*, Presented at ECS 244, Gothenburg, Sweden, 2023. [URL:](https://iopscience.iop.org/article/10.1149/MA2023-02381861mtgabs/meta)
<https://iopscience.iop.org/article/10.1149/MA2023-02381861mtgabs/meta>
- 10 **E.A. Chadwick**, H. Parmar, P. Shrestha, A. Bazylak, and V. Schulz, *Biomimetic microchannels for the passive management of water in pem fuel cells*, Presented at ECS 242, Atlanta, USA, 2022. [URL:](https://iopscience.iop.org/article/10.1149/MA2022-02401455mtgabs/meta)
<https://iopscience.iop.org/article/10.1149/MA2022-02401455mtgabs/meta>
- 11 H. Parmar, **E.A. Chadwick**, P. Shrestha, and A. Bazylak, *Tailoring flow field channel aspect ratio for efficient mass transport and compression in fuel cells*, Presented at ECS 242, Atlanta, USA, 2022. [URL:](https://iopscience.iop.org/article/10.1149/MA2022-02401452mtgabs/meta)
<https://iopscience.iop.org/article/10.1149/MA2022-02401452mtgabs/meta>
- 12 **E.A. Chadwick**, L. Hammen, M. Ioannidis, V. P. Schulz, J. Gostick, and A. Bazylak, *Elucidating the effects of gravity when simulating drainage in porous media via an image-based porosimetry method*, AGU Fall Meeting Abstracts, 2021. [URL:](https://ui.adsabs.harvard.edu/abs/2021AGUFM.H22D..05C/abstract)
<https://ui.adsabs.harvard.edu/abs/2021AGUFM.H22D..05C/abstract>
- 13 J. K. Chan, **E.A. Chadwick**, T. Taniguchi, A. Ahmadipour, T. Suzuki, D. Romero, C. Amon, T. Waddell, G. Karoubi, and A. Bazylak, *Modelling of flow-induced shear stress to predict targeted delivery of cells in the decellularized lung*, Proceedings of The Joint Canadian Society for Mechanical Engineering and CFD Society of Canada International Congress, Abstract ID: 141, 2019.
- 14 **E.A. Chadwick**, T. Suzuki, M. George, D. Romero, C. Amon, T. Waddell, G. Karoubi, and A. Bazylak, *Multiscale fluid flow analysis of decellularized mouse lungs to improve lung regeneration*, Proceedings of The Joint Canadian Society for Mechanical Engineering and CFD Society of Canada International Congress, Abstract ID: 203, 2019.
- 15 **E.A. Chadwick**, T. Suzuki, M. George, D. Romero, C. Amon, T. Waddell, G. Karoubi, and A. Bazylak, *In-silica extraction of mouse lung vasculature via micro-ct for modelling de- and recellularization*, Presented at Biomedical Engineering Society Annual Meeting, Atlanta, USA, Abstract ID: 718, 2018.
- 16 **E.A. Chadwick**, T. Suzuki, M. George, D. Romero, C. Amon, T. Waddell, G. Karoubi, and A. Bazylak, *Micro-ct study of mouse lung vasculature: A foundation for modelling lung de- and recellularization*, Presented at Biomedical Engineering Society Annual Meeting, Phoenix, USA, Abstract ID: 1382, 2017.

Skills

Languages	English (Native), German (A2)
Coding	Python, Matlab, C++, ImageJ Macro Language
Software	Fiji (ImageJ), SolidWorks, AutoDesk CFD, AutoDesk Inventor, GeoDict, Dragonfly, COM-SOL, OriginPro, Inkscape, Microsoft Office
Misc.	Academic research, teaching, training, mentoring

Awards & Scholarships

2024 – 2025  **Ontario Graduate Scholarship**, University of Toronto (\$10,000 CAD)

Awards & Scholarships (continued)

	■ Hatch Graduate Scholarship for Sustainable Energy Research , University of Toronto (\$10,000 CAD)
2023 – 2024	■ Queen Elizabeth II Graduate Scholarship in Science & Technology , University of Toronto (\$15,000 CAD)
2023	■ Climate Positive Energy Scholarship , University of Toronto (\$15,000 CAD)
	■ Mitacs Globalink Research Award , Mitacs (\$6,000 CAD)
2023 – 2024	■ Hatch Graduate Scholarship for Sustainable Energy Research , University of Toronto (\$10,000 CAD)
2022	■ Faculty of Applied Science & Engineering Graduate Student Endowment Fund , University of Toronto (\$6,250 CAD)
	■ Hatch Graduate Scholarship for Sustainable Energy Research , University of Toronto (\$10,000 CAD)
2021 – 2022	■ Queen Elizabeth II Graduate Scholarship in Science & Technology , University of Toronto (\$15,000 CAD)
	■ Ontario Graduate Scholarship , University of Toronto (\$10,000 CAD - Declined due to overlap with Queen Elizabeth Scholarship)
2020 – 2021	■ Ontario Graduate Scholarship , University of Toronto (\$10,000 CAD)
2018 – 2019	■ Ontario Graduate Scholarship , University of Toronto (\$10,000 CAD)
2017	■ Barbara & Frank Milligan Graduate Fellowship , University of Toronto (\$5,460 CAD)
2016	■ Undergraduate Research Assistant Award , University of Guelph (\$6,989 CAD)
2014	■ Undergraduate Research Assistant Award , University of Guelph (\$6,989 CAD)
2013	■ Lindsay W. Christie Bursary , University of Guelph (\$1,000 CAD)

Certifications

2019	■ Certificate of Data Science , SciNet Courses: <i>Neural Networking Programming, Quantitative Applications for Data Analysis</i>
	■ Second Degree Black Belt , Gōjū-ryū Karate
2011	■ First Degree Black Belt , Gōjū-ryū Karate